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Confidence intervals in dermatology

Z.H. Hopkins, C. Moreno and A.M. Secrest

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Summary

This study from the U.S.A. aimed to find out how often confidence intervals are used when reporting dermatology research. Confidence intervals relate to two values ('intervals') that the researchers, using data analysis, believe that the actual, true value will fall within. A 95% confidence interval indicates that if an experiment were repeated 100 times, and a 95% confidence interval was calculated each time, 95 of the intervals would contain the true value while five of the intervals would not contain the true value. A confidence interval might be used when you want to estimate something about a very large number of people, and can therefore only use a sample of those people. For example, if you want to know how many people in the UK have dry skin, you cannot ask every person, so you use a sample and then calculate a range for the mean number of people in your sample who have dry skin. As the sample is never going to be an exact representation of the whole population, you might say that you are 95% confident that the mean number of people in the UK with dry skin is, for example, between 20% and 30%. Confidence intervals help readers understand both the statistical significance and the clinical significance more easily. Most modern statistical software packages conveniently calculate confidence intervals and provide an intuitive way to understand study results. The authors collected dermatology articles from a variety of major dermatology journals as well as dermatology articles from the *New England Journal of Medicine* over the past decade. The authors found that reporting of this important metric is disappointingly low in dermatology journals, with only 1 in 5 articles containing confidence intervals, which was significantly lower than reporting in the *New England Journal of Medicine*. Articles published more recently were more likely to report confidence intervals. After controlling for factors like type of study and year of publication, the *New England Journal of Medicine* still reported confidence intervals more frequently. This is likely due to their extremely rigorous statistical peer review process. While such strict requirements cannot be adopted by all journals, increased awareness of this simple metric could significantly improve the statistical reporting of the dermatology literature. We urge dermatology researchers to include this important metric in their findings. Likewise, we recommend improved statistical training as part of dermatology residency training.